

Task 6 — The Art of Data Storytelling

Data Analyst · Phase 1 Industry Immersion · Study Guide

DATA ANALYST

PlaceMux · Phase 1 · Study Guide

DIFFICULTY

Foundational · Basic

Level 6 of 20 · ramp 29%

1 · What this is & why it matters

Turn analysis into a clear narrative: structure findings so a busy decision-maker grasps them fast and acts. An insight nobody understands changes nothing. Storytelling is what converts analysis into decisions.

2 · Learning objectives

By the end of this guide and task you will be able to:

- Deliver: A short narrative report/deck with a clear takeaway and recommendation.
- Explain the key concepts behind the task in your own words.
- Choose between alternative ways to execute it, and justify your choice.
- Avoid the common pitfalls and know how your work is scored.

3 · Prerequisites

- A reproducible Python data environment (pandas, numpy, matplotlib).
- Comfort reading data into a DataFrame and writing basic SQL.
- The habit of validating data before analysing it.

4 · Key concepts (understand these first)

Audience-first framing

start from the decision the reader must make.

Narrative arc

context -> tension (the problem) -> insight -> recommendation.

Signal vs noise

show the few numbers that matter, hide the rest.

Chart choice

the right chart type for the message (trend=line, comparison=bar, part-of-whole=stacked).

Annotation

the chart should state its own takeaway in words.

5 · Recommended stack & tools

- matplotlib/seaborn
- slides or a notebook report
- clear plain-English writing

6 · The build pipeline (follow in order)

This is the flow from nothing to demoable. Each step builds on the last; the final steps verify and prove the work.

1. Write the one-sentence takeaway before making any slide.
2. Order findings as a story, not as your analysis sequence.

3. Pick the minimum charts that prove the takeaway.
4. Annotate each chart with its point.
5. End with a concrete recommendation and its expected impact.
6. Dry-run the story on someone who wasn't in the analysis.

7 · Other ways to execute (alternative approaches)

There is rarely one right path. Consider these alternatives and pick deliberately for your context:

- Slide deck vs one-page memo vs live dashboard walkthrough.
- Top-down (answer first) vs bottom-up (build to the answer) structure.

8 · Brainstorming & deeper thinking

Sit with these questions — they separate someone who completed the task from someone who understood it:

- If the reader remembers one number, what should it be?
- Where might a sceptical exec push back, and is your evidence ready?
- What action does each chart drive?

9 · Definition of Done

Done when all are true and demoable

- A short narrative report/deck with a clear takeaway and recommendation.
- Demonstrable live on real data.

10 · How your work is evaluated (out of 100)

Scoring parameter	Marks
Core deliverable — A short narrative report/deck with a clear takeaway and recommendation.	50
Real-data quality & correctness (real inputs at realistic scale, not a toy/happy-path)	20
Live verification & evidence (demonstrated live; real output, not claims)	15
Dependency, failure & edge-case handling (errors handled; hand-offs honoured)	15
TOTAL	100

11 · Pitfalls to avoid

Worry if any of these are true of your work

- Leading with methodology instead of the answer.
- Chart junk and too many decimals.
- No recommendation -- analysis that just 'shows' but doesn't 'say'.

12 · Where this fits & go deeper

Over Phase 1 you grow from loading a CSV to shipping a live, decision-grade analytics product. You turn raw data into numbers people trust and decisions people make.

Go deeper (optional further study):

- Statistics for analysts (distributions, tests, intervals)
- SQL window functions and query optimisation
- Storytelling with data and dashboard design
- Time-series and segmentation techniques